

CE802 Worked Solutions

All questions solved with formulas, calculations, and concise explanations.

Where exact fractions are natural, they are kept exact; decimal answers are shown to three decimal places when needed.

Question 1 - Incremental 1-NN retention decisions

Similarity measure: number of identical attributes among {Size, Shape, Colour}. The system tests condition 1 first, then condition 2, then condition 3.

Example	Decision	Why
medium, sphere, red : Poor	Retained because of condition 1	No saved example exists yet, so it has zero similarity to the saved set.
medium, pyramid, brown : Poor	Not retained	Nearest saved example is (medium, sphere, red) with similarity 1 and the predicted class is Poor, so it is classified correctly.
small, sphere, yellow : Good	Retained because of condition 3	Nearest saved example is (medium, sphere, red) with similarity 1 and class Poor, so the new example would be misclassified.
small, sphere, red : Good	Retained because of condition 2	It has two equally similar nearest neighbours with similarity 2: (medium, sphere, red): Poor and (small, sphere, yellow): Good.
medium, sphere, yellow : Poor	Retained because of condition 2	It has two equally similar nearest neighbours with similarity 2: (medium, sphere, red): Poor and (small, sphere, yellow): Good.
large, cube, brown : Good	Retained because of condition 1	Similarity is 0 to every saved example, so condition 1 applies.

Question 1 continued - remaining examples

Example	Decision	Why
small, sphere, brown : Poor	Retained because of condition 3	Nearest neighbours are (small, sphere, yellow): Good and (small, sphere, red): Good, both with similarity 2, so 1-NN predicts Good and misclassifies the example.
large, cube, red : Good	Not retained	Nearest saved example is (large, cube, brown): Good with similarity 2, so it is classified correctly.

Question 2 - Bias-variance idea

Correct answer: (a) overfitting and can be reduced by reducing the complexity of the hypothesis class.

Reason: large variance means the model is fitting noise in the training set; that is the hallmark of overfitting.

Question 3 - Confusion-matrix measures

From the confusion matrix: TP = 45, FN = 8, FP = 5, TN = 64, Total = 122.

Measure	Formula	Substitution	Result
Estimated prior P(Yes)	$P(\text{Yes}) = (TP + FN) / \text{Total}$	$(45 + 8) / 122 = 53 / 122$	0.434
Recall	$\text{Recall} = TP / (TP + FN)$	$45 / (45 + 8) = 45 / 53$	0.849
Precision	$\text{Precision} = TP / (TP + FP)$	$45 / (45 + 5) = 45 / 50$	0.900
Accuracy	$\text{Accuracy} = (TP + TN) / \text{Total}$	$(45 + 64) / 122 = 109 / 122$	0.893

Question 4 - MDP and Bellman optimality

Given: $L = 1/6$, $G = 7$, $\alpha = 1/7$, $\beta = 6/7$, $\lambda = 1/4$, $\delta = 3/4$, and discount factor $\gamma = 1$.

Bellman optimality: $Q^*(s, a) = R(s, a) + \sum_{s'} T(s' | s, a) \max_{a'} Q^*(s', a')$

Requested transition probabilities from s_{12} under action a_0 :

$T(s_{22} | s_{12}, a_0) = \beta = 6/7$

$T(s_{23} | s_{12}, a_0) = \alpha = 1/7$

Terminal cash rewards:

State	Reward formula	Value
s_{31}	L^3	$1/216$
s_{32}	$L^2 G$	$7/36$
s_{33}	$L G^2$	$49/6$
s_{34}	G^3	343

Backward-induction values are shown on the next page.

Question 4 continued - backward induction

Quantity	Calculation	Value
$Q^*(s_{21}, a_0)$	$(6/7)(1/216) + (1/7)(7/36)$	$2/63$
$Q^*(s_{21}, a_1)$	$(1/4)(7/36)$	$7/144$
$V^*(s_{21})$	$\max(2/63, 7/144)$	$7/144$
$Q^*(s_{22}, a_0)$	$(6/7)(7/36) + (1/7)(49/6)$	$4/3$
$Q^*(s_{22}, a_1)$	$(1/4)(49/6)$	$49/24$
$V^*(s_{22})$	$\max(4/3, 49/24)$	$49/24$
$Q^*(s_{23}, a_0)$	$(6/7)(49/6) + (1/7)(343)$	56
$Q^*(s_{23}, a_1)$	$(1/4)(343)$	$343/4$
$V^*(s_{23})$	$\max(56, 343/4)$	$343/4$
$Q^*(s_{11}, a_0)$	$(6/7)(7/144) + (1/7)(49/24)$	$1/3$
$Q^*(s_{11}, a_1)$	$(1/4)(49/24)$	$49/96$
$V^*(s_{11})$	$\max(1/3, 49/96)$	$49/96$
$Q^*(s_{12}, a_0)$	$(6/7)(49/24) + (1/7)(343/4)$	14
$Q^*(s_{12}, a_1)$	$(1/4)(343/4)$	$343/16$
$V^*(s_{12})$	$\max(14, 343/16)$	$343/16$
$Q^*(s_0, a_0)$	$(6/7)(49/96) + (1/7)(343/16)$	$7/2$
$Q^*(s_0, a_1)$	$(1/4)(343/16)$	$343/64$
$Q^*(s_{31}, \text{cash})$	L^3	$1/216$
$Q^*(s_{32}, \text{cash})$	$L^2 G$	$7/36$

Optimal policy: choose a_1 for all states at time 0, 1, and 2, because $Q^*(s, a_1) > Q^*(s, a_0)$ at every decision state.

Question 5 - Interpretation of the optimal policy

The policy is not risk-averse; it is expected-value optimal. Action a_1 has a 75% chance of total loss each year, but its expected one-step multiplier is $E[a_1] = (1/4)*7 = 7/4$, while $E[a_0] = (1/7)*7 + (6/7)*(1/6) = 8/7$. Because $7/4 > 8/7$, Bellman optimality favours a_1 even though it is much riskier.

If the horizon is k years and the optimal policy keeps choosing a_1 , then the probability of receiving any terminal reward (equivalently, surviving all k years) is $(1/4)^k$, because survival requires k consecutive successes. As $k \rightarrow \infty$, this probability tends to 0. This happens because Q^* maximizes expected return, not survival probability or worst-case safety.

To include a survival constraint, modify the reward so ruin is explicitly penalized, for example by assigning a large negative reward $-M$ whenever the process enters state X , or by maximizing a risk-sensitive utility rather than raw final wealth. Then the policy would trade off reward against ruin and would become more conservative, possibly switching to a_0 in some or all states.

Question 6 - k-means clustering (k = 2)

Points: a = (3, -5), b = (-3, -1), c = (-2, 2), d = (0, 1), e = (2, 2), f = (5, -5). Initial centroids: c1 = (-3, -1), c2 = (2, 2).

At each iteration, assign every point to the nearest centroid using Euclidean distance, then recompute each centroid as the mean of the assigned points: $c_k = (1 / |C_k|) \sum_{x \in C_k} x$.

Iteration	Cluster 1	Cluster 2	New c1	New c2
1	{b, c}	{a, d, e, f}	$((-3)+(-2))/2,$ $((-1)+2)/2 = (-2.500,$ $0.500)$	$(3+0+2+5)/4,$ $(-5+1+2-5)/4 =$ $(2.500, -1.750)$
2	{b, c, d}	{a, e, f}	$((-3)+(-2)+0)/3,$ $((-1)+2+1)/3 =$ $(-1.667, 0.667)$	$(3+2+5)/3, (-5+2-$ $5)/3 = (3.333,$ $-2.667)$
3	{b, c, d, e}	{a, f}	$((-3)+(-2)+0+2)/4,$ $((-1)+2+1+2)/4 =$ $(-0.750, 1.000)$	$(3+5)/2, (-5-5)/2 =$ $(4.000, -5.000)$

After iteration 3, the assignments remain the same, so the algorithm has converged.

Therefore the correct membership choices are: Iteration 1 -> cluster 1 = b,c and cluster 2 = a,d,e,f; Iteration 2 -> cluster 1 = b,c,d and cluster 2 = a,e,f; Iteration 3 -> cluster 1 = b,c,d,e and cluster 2 = a,f.

Question 7 - Which clustering was NOT obtained by k-means?

Answer: A.

Justification (within 50 words): k-means produces Voronoi partitions, so with Euclidean distance each cluster must be separated by linear boundaries and is therefore convex. In plot A, each colour wraps around the other in a non-convex way, so that clustering cannot come from k-means.

Question 8 - Reliability issue in the Python code

Main issue: data leakage caused by performing feature selection on the full dataset before cross-validation. SelectKBest is fit once using all examples, so information from the validation folds leaks into the selected feature set.

Implication: the reported cross-validation accuracy is optimistically biased. The model is evaluated on validation folds that have already influenced the chosen features, so the score is no longer a trustworthy estimate of generalization performance.

Fix: place feature selection inside a Pipeline and cross-validate the whole pipeline, so each training fold performs its own SelectKBest fit using training data only. If k or max_depth must be tuned, use nested cross-validation (inner loop for model selection, outer loop for unbiased evaluation). A stratified split is also advisable for classification so class proportions are preserved across folds.

```
Correct pattern: Pipeline([('selector', SelectKBest(...)), ('model',  
DecisionTreeClassifier(...))])
```

End of worked solutions